PlanoVision - Smart Integrated Event Planning and Business Intelligence System

Vendor Intelligence & Smart Package Recommendation Engine

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Project Proposal Report

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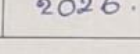
September 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

For the successful organization of an event, it is important to choose some vendors who offer different types of services including catering, photography, decorations, entertainment and sound & visuals. At present, these vendors are generally chosen individually depending upon different considerations like cost, rating, availability and type of service and it is not easy to take care of all the issues that can come up with the use of multiple vendors. This can lead to problems such as time consumption, scheduling problems, unexpected conflicts and additional costs.

The individual research problem of this research is the lack of identifying several compatible vendors according to the event requirements and budget and recommending an appropriate vendor package considering venue suitability and operational risks. Although existing vendor platforms often provide facilities for individual vendor search, filtering and comparison, the literature and existing systems analysis identified that the integration of multi-vendor compatibility, venue suitability, risk analysis and explainable package-level recommendation into a single decision process is limited.

To address this gap, a Vendor Intelligence and Smart Package Recommendation component is proposed. It combines compatible vendors based on event requirements, budget, vendor reliability, availability, compatibility, operational risks, and venue suitability to recommend a Smart Vendor Package, and explains the reasons for the recommendation.

This research is conducted on Design Science Research (DSR) methodology, and the development uses Agile Development Approach with Iterative Development. The system is evaluated on recommendation accuracy, vendor compatibility, venue suitability, conflict identification, explanation quality, response time, and usability, compared to the Criteria-Based Individual Vendor Selection baseline. This component acts as a key decision-support function for intelligent vendor selection and package recommendation in a complete Integrated Event Management System.

Keywords: Vendor Recommendation, Vendor Intelligence, Multi-Criteria Decision Making, Explainable AI, Event Management

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
DSR	Design Science Research
MCDM	Multi-Criteria Decision Making
ML	Machine Learning
SUS	System Usability Scale
UI	User Interface
UAT	User Acceptance Testing
WBS	Work Breakdown Structure
XAI	Explainable Artificial Intelligence

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1 INTRODUCTION

Background and Context

When planning an event, several vendors must be selected for various services such as catering, photography, and decoration Currently, vendors are often selected separately. As a result, planners may find it difficult to determine whether the selected vendors are compatible, available, suitable for the venue, and free from operational conflicts. As a result, event planners must check these factors manually, which can lead to problems such as time-consuming, conflicts between vendors, and unexpected problems. Therefore, a more efficient method is needed to select a complete vendor package consisting of several suitable vendors by considering factors such as event requirements, budget, vendor reliability, compatibility, availability, and venue suitability. The problem at hand is affecting the event planners, the event organizers and the clients. First, the event planners must manually check things like price, availability, reliability, compatibility and the venue the different vendors offer. Also, there is a risk of high costs, scheduling problems, conflicts between vendors and unexpected issues due to making wrong decisions in vendor selection. Therefore, selecting vendors who are suitable and compatible with each other for an event is an important part of the event planning process.

The event industry also makes an important contribution to the global economy. According to the Events Industry Council and Oxford Economics, about 1.65 billion people will have participated in business events worldwide by 2025, generating about 1.3 trillion US dollars in direct spending [1]. MICE tourism in Sri Lanka also makes an important contribution to the tourism sector. According to the Sri Lanka Tourism Development Authority, MICE tourism represented about 11% of the tourism sector in Sri Lanka between 2015 and 2019 and has provided an average revenue of about 160 million US dollars per year [2]. These statistics illustrate the importance of event-related activities in Sri Lanka as well as in the world. Therefore, it is important to provide technological solutions to streamline the event planning process and select suitable vendors.

Currently, various online platforms are used to find and select vendors for event planning. The Cvent Vendor Marketplace uses AI to find suitable vendors and venues based on event details, compare quotes, and manage vendor information [3]. The Knot Vendor Marketplace enables wedding vendors to be found and compared based on factors such as location, price, services, availability, and customer reviews [4]. However, while these systems provide vendor search, matching, and comparison, it is limited in the ability to fully consider whether multiple vendors are compatible with each other, what their operational risks are, and whether their services and equipment are suitable for the selected venue in a single decision-making process. There is also a need to recommend a complete multi-vendor package that considers factors such as budget, compatibility, reliability, and venue suitability, and to explain the rationale for that recommendation.

As a solution to this need, a system is proposed that identifies suitable vendors according to event requirements, considers their reliability, compatibility, availability, venue suitability and budget, and recommends a complete multi-vendor package. The recommendation is also explained to the user using Explainable AI (XAI) to make the reasons for the package selection easy to understand. The main components of this proposed solution include Vendor Intelligence, Vendor Compatibility and Risk Analysis, Venue Suitability Validation, Multi-Criteria Recommendation, Smart Package Generation and Explainable AI (XAI). By working together, these components enable the system to identify vendors that are suitable for the event and provide a suitable complete vendor package.

This research is related to Sustainable Development Goal 9 (SDG 9) – Industry, Innovation and Infrastructure. SDG 9 focuses on making industry and service processes more efficient and improved using technology and innovation. This research component uses AI and data-driven methods to analyze vendor compatibility, operational risks, and venue suitability and recommend a suitable multi-vendor package. This research contributes to improving the process of manually finding and comparing vendors in a more efficient and technological way.

This research is also linked to the SST – Software Systems and Technologies research cluster. This research component focuses on developing a software-based intelligent system to analyze event requirements and vendor data, select suitable vendors, and recommend a complete vendor package. Here, software technologies such as Artificial Intelligence, data analysis, recommendation techniques, and Explainable AI (XAI) are used to analyze vendor reliability, compatibility, operational risks, and venue suitability. This will enable an intelligent decision-support software solution for the event planning process. Accordingly, this research, which focuses on developing an intelligent decision-support system using software technologies, is directly aligned with the SST – Software Systems and Technologies research cluster.

1.1 Literature Review

In event management, vendor selection is a decision-making problem that is not limited to selecting one vendor, but several criteria must be considered at the same time. While selecting several vendors for various services such as catering, photography, decoration, entertainment, audiovisual services, factors such as price, service quality, reliability, availability and suitability should be considered. Therefore, the vendor selection problem is related to the research areas related to Multi-Criteria Decision Making (MCDM) and Artificial Intelligence (AI).

Ho, Xu and Dey (2010) conducted a literature review on supplier evaluation and selection and analyzed 78 journal articles published between 2000–2008 [1]. In this research, various supplier-selection methods and evaluation criteria have been studied and it has been shown that not only cost but also several criteria such as quality, delivery performance and

reliability are required for supplier selection. The method of this studatasets systematic literature review, and instead of a separate experimental dataset, previous peer-reviewed research publications have been used as their evidence base. Its main strength is to collect a large amount of research on supplier selection and identify common evaluation criteria and decision-making approaches. But its context is general supply-chain supplier selection and selection of several event vendors as one package, less attention is paid to vendor compatibility or event venue suitability. According to these findings, it can be decided that the vendor selection of the proposed system should not be based on a single criterion, but based on several criteria such as price, rating, reliability, availability, compatibility and venue suitability [1].

Chai, Liu and Ngai (2013) conducted a systematic review of decision-making techniques used in supplier selection and analyzed 123 journal articles [6]. They have classified supplier-selection methods under major categories such as MCDM, mathematical programming and artificial intelligence. The method here is a systematic literature review, and previous supplier-selection studies have been used as the dataset. Research has shown that it is important to choose decision-making techniques or combine several techniques according to the nature of the supplier-selection problem. The strength of this research is the systematic comparison of different decision-making methods used for supplier selection. But the research context focuses more on traditional supplier-selection problems and does not cover issues such as event-specific vendor compatibility, venue constraints and package-level recommendations. Therefore, these findings provide a methodological foundation for the complementary use of MCDM and AI techniques in the proposed research [2].

Keshavarz Ghorabaee et al. (2017) conducted a review of Multi-Attribute Decision-Making methods used for supplier evaluation and selection in uncertain and fuzzy environments and analyzed 339 publications [7]. In this research, AHP, TOPSIS and other MCDM approaches have been considered for supplier-selection problems. Its context is supplier-selection environments with uncertainty and subjective judgment. Findings indicate the importance of handling multiple criteria and uncertain information through structured decision-making techniques in supplier evaluation. The strength of this study is the analysis of MCDM methods on a large publication base. But relationships such as compatibility between the event management context and vendors are not considered there. The findings are important for the criteria weighing and vendor ranking process of the proposed system and guide the determination of vendor suitability through a structured multi-criteria approach [3].

Research has also been done on making supplier selection more data-driven using AI and Machine Learning. Chai and Ngai (2020) have reviewed recent developments in supplier-selection decision-making techniques and pointed out the development of hybrid decision-making methods, big data and AI-based approaches [8]. The method of this research is to review recent supplier-selection literature, and it shows the relationship between traditional MCDM methods and modern data-driven techniques in supplier-selection research. Its strength is in explaining how AI and data analytics can be used for supplier selection. But its research context is general supplier selection and does not present a complete package

recommendation for event vendors. These findings are important to the proposed Vendor Intelligence component and support that AI and data-driven methods can be used to analyze vendor data and identify relevant vendor characteristics [8].

In research published in Decision Analytics Journal in 2023, a decision support system was presented using a Random Forest-based Machine Learning approach to classify supplier-selection criteria [9]. In this research, 30 potential criteria for supplier selection were considered and critical criteria were identified using Random Forest and feature-selection techniques. Criteria such as quality, on-time delivery, material price and information sharing have been identified as important, and it has been shown that removing non-critical criteria increased classification accuracy and F-score. The method here is based on supervised machine learning and feature selection and the supplier-selection criteria data has been used for training and evaluation. Its strength is to identify the criteria's importance as data-driven rather than based on purely manual assumptions. But this pays more attention to individual supplier criteria classification and does not recommend a combination of several vendors or an event package. These findings affect the proposed Vendor Intelligence component by indicating that relevant features can be identified from vendor data and used in the recommendation process [9].

Recent research on explainable supplier selection combining Machine Learning and MCDM has presented a hybrid framework [6]. In this approach, complex supplier data is analyzed by Machine Learning and supplier ranking is done by MCDM techniques. The research has evaluated the framework using real-world manufacturing-related case studies and has shown the ability to combine intelligent data analysis and structured decision-making. Its main strength is providing a balance between Machine Learning performance and human-understandable decision-making. That way, not only the result obtained from the system, but also the criteria that affect the decision can be explained. However, the research context is manufacturing supplier selection and does not include event vendor compatibility, venue suitability and complete event package generation. Therefore, this research provides a major methodological influence to develop an integrated decision process such as AI → MCDM → ranking → explanation in the proposed system [6].

There is also research that considers vendor reliability and operational risk as important factors in the supplier-selection decision. In a recent risk-informed supplier-selection research, supplier delivery delays were predicted using machine learning, and those predictions were integrated for supplier-selection and allocation decisions [7]. This study has been evaluated using an automotive-sector case study and it has been shown that supplier-related future risks can be included in the decision-making process through predictive analysis. The strength here is to consider not only the supplier's historical performance but also the potential future risk. But the risk context here is mainly around supply-chain delivery delays and does not cover factors such as vendor availability, late arrival, scheduling conflict, service capacity and equipment-related risks in event management. Therefore, this concept can be adapted for event-specific operational risks in the proposed Vendor Compatibility and Risk Analysis component [7].

Vultureanu-Albişi and Bădică (2022) conducted survey research on the explainability of recommendation systems and analyzed the effect of adding explanations to recommender systems on user understanding and confidence [12]. The method of this research is a survey of existing explainable recommender-system literature, and previous studies on recommendation explanations have been used as evidence. Instead of giving only the recommendation according to the findings, it is important to explain the reasons for the recommendation to the user. The strength of this study is the consideration of human-centered factors such as user understanding and trust in addition to technical recommendation performance. This gives a direct influence to the Explainable AI component of the proposed system. For example, when the system recommends a vendor package, it can be explained to the user that it was selected due to factors such as budget fit, vendor reliability, availability, vendor compatibility and venue suitability [12].

As an event-specific recommendation research, Horowitz, Contreras and Salamó (2018) presented a mobile recommender system called EventAware [13]. This system recommends events to users using contextual and tag-based recommendation techniques. LinkedIn profiles for user-related information and tags obtained from online sources for event information have been used. In the research, challenges such as changing user preferences, limited feedback for future events and shorter event availability period have been considered. Its strength is to include event context and user information in the recommendation process. But the recommendation target is the event, and its purpose is not to recommend the vendors needed to organize an event. Therefore, EventAware's contextual recommendation concept influences the proposed system's event type, event requirements and user preferences to be used as input to the vendor selection process [13].

For venue recommendation, Zhang et al. (2019) presented a group event venue recommendation system called GEVR [14]. In this research, a real-world dataset obtained from the OutWithFriendz application was used and included 625 users and more than 500 group events. The system combines a social-based group location prediction technique and a Foursquare Venue Recommendation API. According to research findings, more than 80% accuracy has been obtained for group meeting-location clusters, and it has been reported that the performance of the proposed approach is better than comparison approaches. The strength here is the use of real-world event data and group context. But the recommendation target of the system is the venue, and it does not evaluate whether the vendor's services and equipment requirements match the selected venue. Therefore, this research provides a basis for the proposed Venue Suitability Validation component and can be adapted to consider venue characteristics and vendor requirements [14].

In addition to research from academic sources, there is also a commercial system for vendor selection in events. Cvent Vendor Marketplace is a commercial application that helps event planners in vendor and venue selection [11]. This application uses Artificial Intelligence for the analysis of information such as event location, dates, number of people, budget and requirements, and then matches the appropriate vendors and venues. Categories of vendors include venues, catering, transportation, entertainment, decoration, staffing, gifts and audiovisuals. It also provides functions for comparing vendor quotes and getting AI-assisted

summaries. Its target users are corporate event planners, third-party event agencies, destination management companies, travel advisors, associations and marketing teams. Cvent's main strength is AI-assisted vendor matching and centralized vendor sourcing. But its publicly described functionality is mainly around vendor sourcing, matching and comparison, and a mechanism that evaluates several vendors' compatibility, event-specific operational risk, detailed venue suitability and complete explainable package generation at the same time has not been clearly specified [11]. Therefore, the proposed research goes beyond vendor-matching capability like Cvent and focuses on providing package-level intelligent decision support.

The Knot Vendor Marketplace is a commercial platform used to find and select wedding vendors [12]. Users can search vendors by location, vendor category, price and preferences and check reviews, photos, pricing and service information. The platform also supports vendor comparison, communication and booking functions. And The Knot's Upload List feature uses AI to analyze uploaded PDF or JPG vendor lists and supports identifying matching vendors in the marketplace. Its strength is providing a centralized marketplace for wedding vendors and search, filtering, comparison, review and booking facilities. But its described functions are mainly around individual vendor discovery and selection, and whether it is compatible with multiple vendors, suitable for the event venue, whether there are operational risks and what is the overall suitability as a complete package is not clearly stated [12]. Therefore, the proposed Smart Package Generation component focuses on generating a coordinated combination of several vendors instead of individual vendor selection.

Considering the above literature and existing systems, several important points can be identified for the proposed research. MCDM-based supplier-selection research points to the need to consider several criteria for vendor selection [1][2][3]. AI and Machine Learning research shows that it is possible to analyze vendor data and identify important features and patterns [4][5]. Hybrid AI and MCDM research show the ability to combine intelligent analysis, structured ranking and explainability [6]. Risk-aware supplier-selection research points to the importance of incorporating reliability and future operational risks into the decision-making process [7]. Explainable recommender research points to the need to explain the reasons for the recommendation to the user [8]. Event and venue recommendation studies confirm that event context and venue information are important to the recommendation process [9][10]. Commercial platforms such as Cvent and The Knot indicate that capabilities such as AI-assisted vendor matching, filtering, comparison and booking are already being used in practical systems [11][12].

But these research and systems separately address specific areas such as individual supplier selection, vendor matching, event recommendation, venue recommendation, risk analysis or explainable recommendation. When selecting catering, photography, decoration, entertainment and several other vendors as one package for an event, it is necessary to simultaneously consider the compatibility, availability, reliability, operational risk and suitability of the vendors with the selected venue. Such an integrated package-level decision-making approach is limited in the literature and existing commercial systems.

Accordingly, this research proposes to address that gap through Vendor Intelligence and Smart Package Recommendation System. First, the Vendor Intelligence component analyzes vendor profiles, ratings, reliability, availability and other relevant information. Then the Vendor Compatibility and Risk Analysis component identifies potential conflicts and operational risks between vendors. The Venue Suitability Verification module ensures that the vendor's offerings and facility requirements fit with the selected venue. The Recommendation module ranks the vendors based on multiple criteria including price, rating, reliability, availability, compatibility, and venue suitability. Instead of recommending individual vendors separately, the Smart Package Generation component generates a complete vendor combination that matches the event requirements. Finally, the Explainable AI component explains to the user the main factors that influenced the final recommendation.

The main purpose of this integrated approach is not to completely replace the existing vendor-search and comparison approaches, but to provide an event-specific, compatibility-aware, risk-aware, venue-aware and explainable vendor package recommendation using vendor information and existing recommendation capabilities. By doing so, event planners and organizers can reduce the time spent manually checking vendors separately and provide a more systematic and transparent decision-making process.

1.2 Research Gap

Although there have been significant advances in techniques such as Multi-Criteria Decision Making (MCDM), Machine Learning and Explainable AI for vendor selection and recommendation in the field of event management, several important issues have not yet been adequately resolved in using those technologies in real event planning environments.

First, it is not enough to make decisions based on only one criterion in the vendor selection, and it is necessary to consider several criteria such as price, quality, reliability, availability and service performance at the same time. Ho et al. [5] and Chai et al. Literature reviews conducted by [2] indicate that MCDM and other decision-making techniques are widely used for supplier selection, while Keshavarz Ghorabaee et al. [7] also points out the need to perform supplier evaluation under uncertainty and fuzzy information. However, this research often focuses on general supplier selection environments and does not fully address the event-specific requirements that arise when selecting multiple vendors for an event as a package.

Secondly, by considering only the individual performance of a vendor, it is difficult to identify operational problems that may arise during an event. For example, facilities matching between a catering vendor and a venue, space required by a decoration vendor, photography vendor's requirements, entertainment equipment and available time slots depend on the relationship between several vendors. Although Abdulla and Baryannis [10] presented an explainable supplier selection approach combining Machine Learning and

MCDM, it has been applied to supplier selection contexts such as oil and gas and aerospace manufacturing. Also, research on risk-informed supplier selection [11] shows the ability to predict risks such as supplier delivery risks, but the consideration of compatibility between event vendors and event-specific operational risks is still limited.

Third, although there is research on event recommendations and venue recommendations, they often focus on choosing an event or a venue separately. Horowitz et al. [9] EventAware recommends events using user preferences and contextual information, while Zhang et al. [10] shows that GEVR recommends suitable event venues for group users. But an integrated approach has not been sufficiently presented to validate a vendor package that matches the venue's physical constraints and facilities vendor requirements.

Fourth, commercial systems such as Cvent Vendor Marketplace and The Knot Vendor Marketplace provide facilities such as vendor search, filtering, reviews, pricing, availability, comparison and AI-assisted matching. Therefore, it cannot be said that there is no AI or automated vendor matching in the existing systems. However, these systems have not adequately addressed vendor-to-vendor compatibility, operational risk, venue suitability and package-level constraints when recommending several vendors together as a package and providing an explainable final recommendation.

Fifth, research on the use of explainable AI for recommendation systems has shown the importance of not only providing a recommendation but explaining the reason for that recommendation to the user. The survey conducted by Vultureanu-Albişi and Bădică [8] shows that explanations can improve users' understanding, confidence and recommendation acceptance. But in an event vendor package recommendation, "Why did you recommend these vendors?", "How does this package fit into the budget?" and "Why are these vendors a good fit for the venue?" The focus on connecting and explaining such things is limited.

Accordingly, the main research gap in this research is that based on event requirements and venue constraints, there is not enough of an integrated approach to recommend a suitable and explainable event package by considering the criteria of vendor price, reliability, availability, compatibility, operational risk and venue suitability at the same time.

To address this research gap, this individual component combines Vendor Intelligence, Vendor Compatibility and Risk Analysis, Venue Suitability Validation, Multi-Criteria Recommendation, Smart Package Generation and Explainable AI. In this way, according to the requirements given by the event planner or event organizer, compatible vendors are identified, venue constraints and operational risks are checked, a suitable vendor package is recommended considering the budget and other criteria, and the reasons for the recommendation are clearly presented.

2 OBJECTIVE

2.1 Main Objective

When the details of the event are given, the system not only looks at the price, but also looks at the reliability, availability, compatibility, risk and venue suitability of the vendors and recommends suitable vendors as a package, explaining why that package was recommended.

2.2 Specific Objectives

- **COLLECTION AND PROCESSING OF DATA:**
  Gather and analyze the event's requirements, vendors' information (fees, ratings, availability times, reliability) and venue information.

- **Compatibility between Vendors and Identification of Risks:**
  Check if the selected vendors can work successfully with each other (Compatibility) and early identification of possible problems between their services and schedules as well as operational risks (Operational Risks).

- **Venue Validation:**
  Verifying through Digital Twin technology whether the space and facilities of the event venue, the equipment of the selected vendors and the preparation requirements are suitable.

- **Automatic recommendation of suitable Vendor Package:**
  Automatically create the most suitable Vendor Package through Multi-Criteria Decision Making (MCDM) methodology considering all the factors such as event budget, requirements, vendors' reliability, availability, compatibility and location suitability.

- **Providing explanations (Explainable AI - XAI):**
  Explaining in natural language the reasons for recommending a certain Vendor Package and why it is the most suitable for the Event so that the user can easily understand.

- **System Performance Evaluation (Evaluation):**
  Evaluating the success of the developed system based on metrics such as accuracy of recommendations, compatibility of vendors, suitability of the venue, response time of the system and user satisfaction.

3 METHODOLOGY

3.1 Research Methodology and Justification

Design Science Research (DSR) methodology has been chosen for this research. The main objective of this research is to identify vendors suitable for an event and develop a software-based solution that can recommend an appropriate vendor package by considering factors such as compatibility, risk, venue suitability, budget and event requirements. Therefore, the DSR methodology is suitable for this research since it is necessary to identify the existing problem, create a practical solution to it, and test and evaluate that solution.

Given the nature of the research problem, this research mainly focuses on creating a software solution and testing its results. The practical problems that arise when selecting vendors for an event are identified and an intelligent vendor recommendation component is created as a solution to them. Therefore, the steps of identifying the problem, developing the solution and evaluating how it works are required.

The data used is structured data such as event requirements, vendor details, pricing, ratings, availability, reliability, vendor compatibility, and venue information. This data is used to identify vendors that are suitable for the event, examine potential conflicts and risks between vendors, determine whether the venue is suitable for the vendors, and finally recommend a suitable vendor package. Therefore, in this research, it is necessary to design, implement, and evaluate the solution along with data analysis.

When considering previous studies related to the research, it is found that methods such as Multi-Criteria Decision Making (MCDM), Machine Learning, and Explainable AI (XAI) have been used for vendor evaluation and selection in research related to supplier selection and recommendation [1–8]. Also, in research related to event recommendation and venue recommendation, it has been shown that recommendations can be given by considering event requirements and venue-related factors [9,10]. The ideas obtained from this research are used to design the vendor evaluation, recommendation, and explanation process of the proposed solution. However, further attention should be paid to providing a package-level recommendation by considering multiple vendors simultaneously, along with compatibility, operational risks, and venue suitability.

To test the research, the system is tested using various event scenarios, vendor combinations, and venue conditions. The results of the system are evaluated on factors such as the accuracy of the recommendations provided, the identification of compatibility between vendors, the examination of venue suitability, the identification of conflicts and risks, and the quality of the explanation provided for the recommendation. This allows us to test how well the proposed solution fits the real event planning needs.

Accordingly, the nature of this research is to create and test a practical software solution, use structured vendor and event data, evaluate the solution through various test scenarios, and use recommendation and decision-making techniques obtained from previous research.

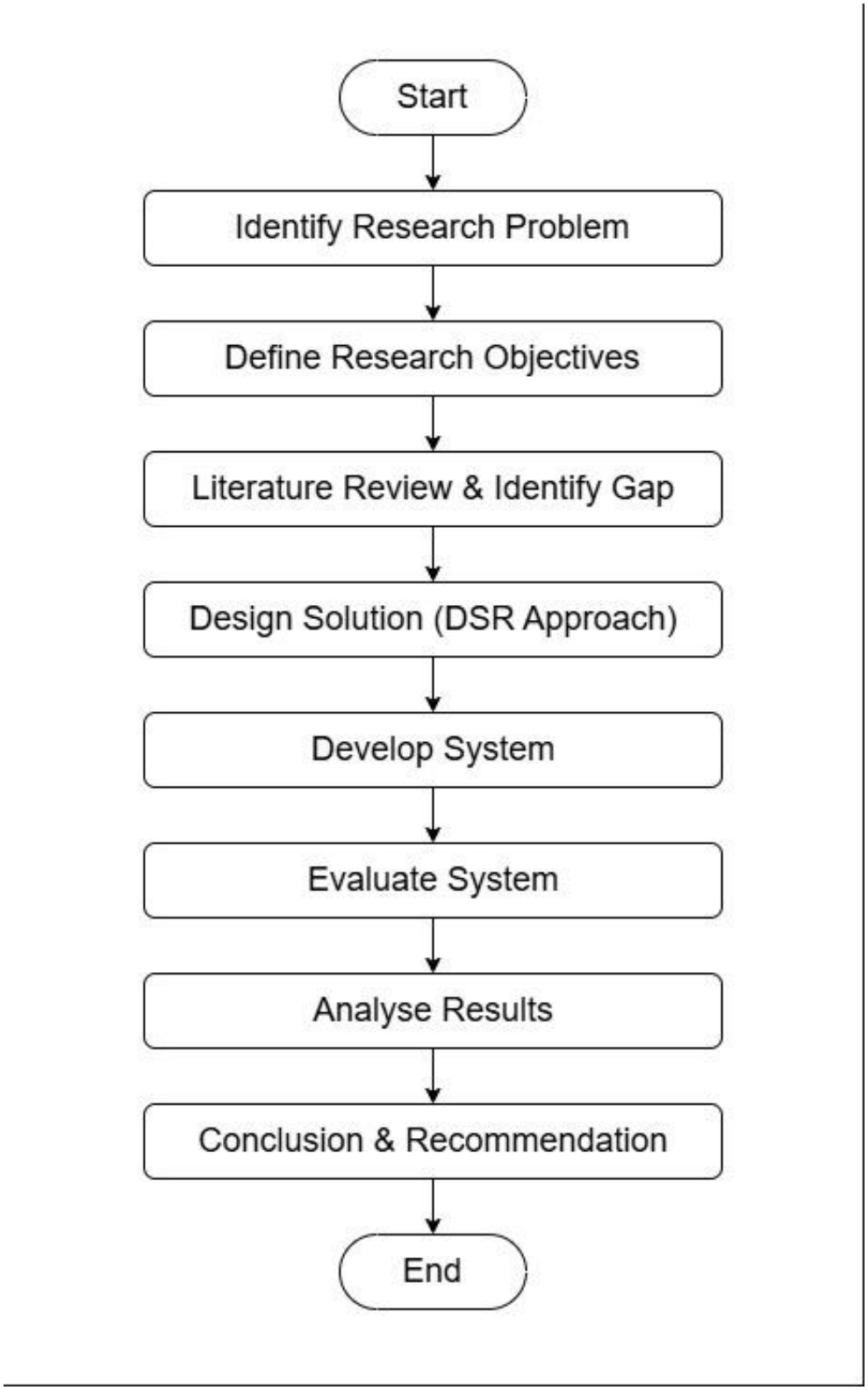


Figure 3.1 Research Methodology

3.2 Data, Participants, and Experimental Inputs

This research mainly uses event information, vendor information and venue information. This data has been selected because this information is necessary to identify suitable vendors according to the needs of the event, examine compatibility issues and risks that may arise between vendors, check whether the venue is suitable for the selected vendors and finally recommend a suitable vendor package.

3.2.1 Data Source

Vendor and venue data are mainly obtained from publicly available websites, vendor profiles, business listings and online sources related to the event industry. Vendor data includes vendors providing services such as Catering, Photography, Decoration, Entertainment and Audio/Visual. Venue data include information such as venue capacity, available facilities, location, available space and facilities required to hold an event.

3.2.2 Sample

Vendor records and event venues belonging to various categories are selected as the sample of this research. The vendor sample includes vendors with different services, price ranges, ratings, availability, and reliability levels, while the venue sample includes venues with different capacities and facilities. Having such different data allows the system to test its recommendations under different event requirements.

3.2.3 Data Sets

There are two sets of data prepared from the collected data, which are vendor data set and venue data set. In the vendor data set, there are attributes like vendor category, services, price range, rating, availability, reliability, and resource requirements. Venue data set includes details like capacity, facilities, location, available space, and facility requirements for vendors.

3.2.4 Participants

A limited group of people with knowledge of Event Planning or Event Management are invited to participate in evaluating the usability and recommendation explanations of the system. They are shown various sample event requirements and the vendor package recommendations provided by the system, and feedback is obtained on whether the recommendations are clear, easy to understand, and useful for decision making.

3.2.5 Data Collection Method

Vendor and venue information is collected from publicly available online sources. A questionnaire or structured evaluation form is used to obtain user feedback. Sample test scenarios are prepared to test the system, including different event types, budgets, guest counts, required services, and vendor requirements. Compatible and incompatible combinations of vendors, and suitable and unsuitable venue conditions are also included for testing.

3.2.6 Data Preparation

Before entering data into the system, the collected data is checked and duplicate records are removed. Missing information is identified and incomplete or incorrect information is corrected where possible. Price, rating, and other numerical values are formatted in a uniform format, and vendor categories and services are also categorized in a uniform manner. The data thus prepared is used for vendor analysis, compatibility checking, venue suitability validation, and vendor package recommendation in the system.

These data, participants, and experimental inputs are selected because they provide the information needed to test the main objectives of the research. By using different event requirements, vendor combinations, and venue conditions, the system can identify suitable vendors, identify vendor conflicts and risks, check venue suitability, and recommend a vendor package that is compatible with the event.

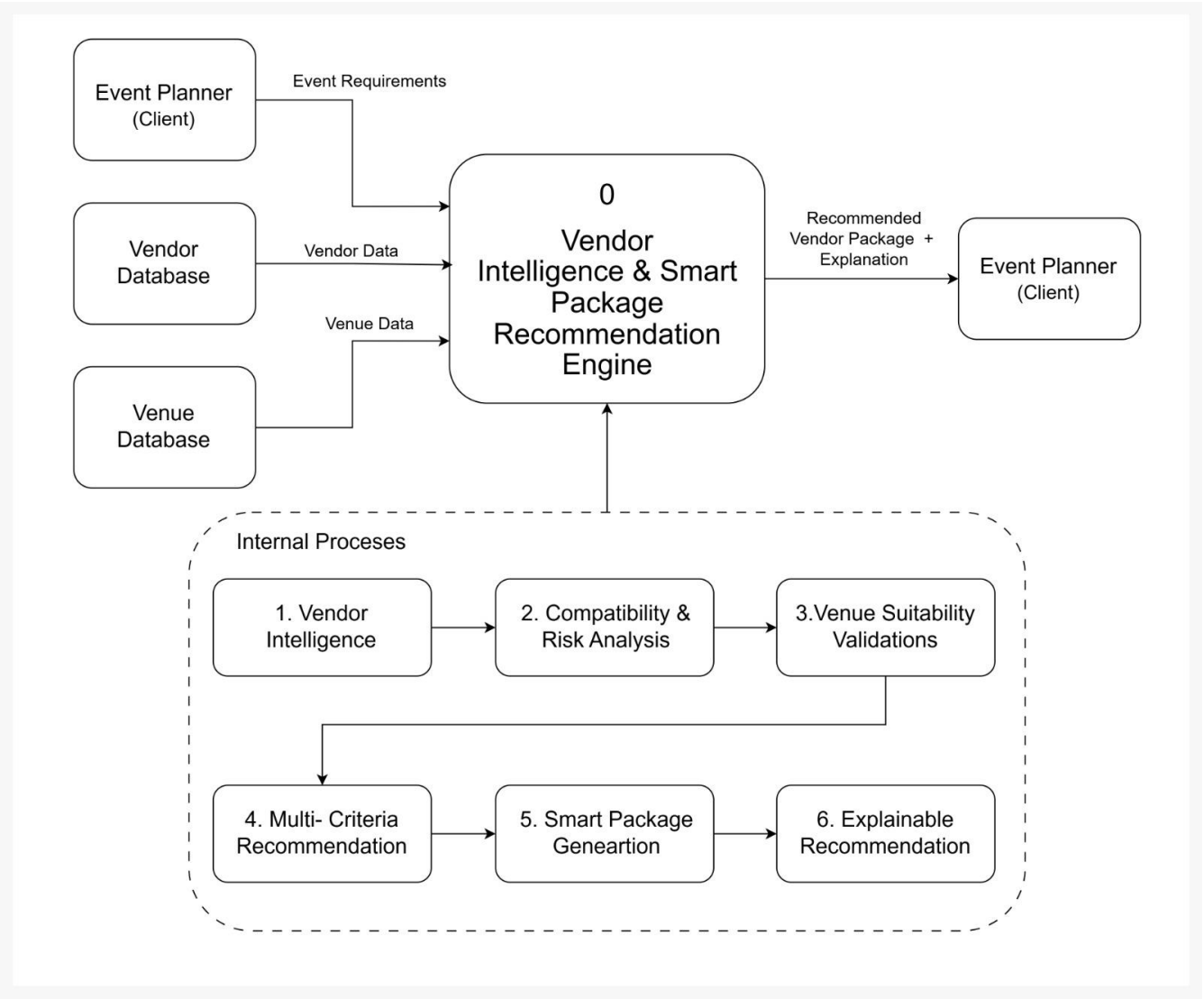


Figure 3.2: Data Flow Diagram

3.3 Baseline / Benchmark

To thoroughly test the performance of the proposed system, the results of the proposed system are compared with the results of the current simple vendor selection method.

In the current vendor selection, vendors can usually be selected by considering factors such as price, rating, availability, and service type separately. It is not possible to fully consider the factors such as contacting participating vendors, whether they are suitable for the position of the event indicator, and whether there are problems in selecting multiple vendors.

Therefore, in this research, the criteria-based individual vendor selection method is used as the baseline. According to the needs of this event, each vendor is selected separately based on factors such as price, rating, availability, and service requirements.

Then, the vendor selection results obtained from that baseline method are compared with the Smart Vendor Package recommendations obtained from the proposed system. It is examined whether the proposed system can provide more suitable and practical sales packages than the basic one, due to additional considerations such as vendor compatibility, location suitability, risk and conflict identification.

Accordingly, the criterion-based individual vendor selection method has been chosen as the baseline/benchmark in this research.

3.4 Evaluation Metrics

Several metrics are used to test whether the proposed system works well. Here, it is tested whether the system correctly selects a vendor package that is compatible with the event, whether the selected vendors are compatible with each other, whether the venue is suitable for the event, whether potential problems are identified, and whether the user can clearly understand the reason for the recommendation provided.

Metric	Why is this tested?	Measurement / Comparison
Recommendation Accuracy	To test whether the system selects the correct vendor package according to the needs of the event.	The recommendations provided by the system are compared with the recommendations obtained from the selected baseline.
Vendor Compatibility	To test whether the vendors selected for the package are compatible with each other and whether there are any problems when working together.	It is tested whether the system correctly identifies compatible and incompatible vendor combinations.
Venue Suitability	To check whether the venue, space, facilities and other requirements are sufficient for the selected vendors.	The results provided by the system are compared with the correct results using suitable and unsuitable venue scenarios.
Conflict Identification	To check whether the system can identify problems that may arise when multiple vendors work on the same event.	The number of problems that the system correctly identifies is tested given previously identified problem scenarios.
Explanation Quality	To check whether the system clearly understands why the system selected a	The explanation provided by users is tested to see if it is

	particular vendor package and why it is suitable for the event.	easy to understand and useful.
Response Time	To check how long it takes the system to recommend a vendor package to the user.	The time it takes to make a recommendation is measured using various test scenarios.
SUS Score	To check whether the system is easy to use for the user	The usability of the system is evaluated using scores from a questionnaire on the usability of the system obtained from users.

Table 3.4 Evaluation Metrics Table

These measures were chosen because they help to test the key functions of the proposed system. The accuracy of the recommendation can be used to test whether an appropriate vendor package is provided, the compatibility between vendors and the identification of problems can be used to identify problems that may arise when working with vendors. The suitability of the venue can be used to test whether the venue is suitable for the event and the selected vendors, and the quality of the explanation can be used to test whether the user can understand the reason for the recommendation. Also, the response time and ease of use of the system can be used to test whether it is suitable for real-world use.

3.5 Validation Strategy

The system is tested in several stages to ensure that the vendor package recommendations provided by the proposed system are accurate, suitable for the event, and meet the objectives of the research.

First, test cases are prepared that include different event types, budgets, guest counts, required services, vendor details, and venue details. These tests include cases where vendors are compatible and incompatible with each other, as well as cases where the venue is suitable and unsuitable. This allows us to test whether the system can make correct decisions under various conditions.

Then, each prepared test scenario is given to the system, and the vendors selected by the system, the vendor package created, the suitability of the venue, the conflicts identified, and the reasons for giving the recommendation are recorded. These results are compared with the predetermined correct results to check whether the system's decisions are correct.

Metrics such as Recommendation Accuracy, Vendor Compatibility, Venue Suitability, Conflict Identification, Explanation Quality, and Response Time are used to measure the

performance of the system. Recommendation Accuracy checks whether the appropriate vendor package is selected according to the needs and budget of the event. Vendor Compatibility checks whether the selected vendors are compatible with each other and whether there are any problems when working together. Venue Suitability checks whether the venue space, facilities, and other requirements are sufficient for the event and the selected vendors. Conflict Identification checks whether the potential problems and operational conflicts between vendors are accurately identified.

Also, Explanation Quality evaluates whether the user can clearly understand why the system selected it and why the package is suitable for the event when it makes the recommendation. Response Time measures the time it takes to make the recommendation. System Usability Scale (SUS) questionnaire is also used to collect feedback from users to check whether the system is easy to use.

In determining the success of the system, the main considerations are providing correct vendor recommendations in test scenarios, selecting suitable vendors, correctly identifying the suitability of the venue, and identifying potential conflicts. Furthermore, it is considered that the explanation provided for the recommendation is clear and useful to users, and that the system provides results in a practical time. The ease of use of the system is also evaluated through SUS results.

Finally, the results of the proposed system are compared with the results obtained from the baseline method selected as Criteria-Based Individual Vendor Selection. In the baseline method, vendors are usually selected separately based on criteria such as price, rating, availability, and service type, and compatibility, venue suitability, and package-level conflicts between vendors are not fully considered. Since the proposed system considers these factors together and recommends a vendor package, it is tested whether it provides more appropriate and practical package recommendations compared to the baseline method.

This validation process evaluates the key aspects of the proposed system: accuracy, compatibility between vendors, venue suitability, conflict detection, clarity of recommendation, response time, and ease of use. Accordingly, the system's functionality and its usefulness for event vendor package recommendation can be clearly confirmed.

3.6 Development Approach

It is appropriate to use the Agile Development Approach, especially the Iterative Development method, for this project. In this system, several parts such as Vendor Intelligence, Vendor Compatibility and Risk Analysis, Venue Suitability Validation, Multi-Criteria Recommendation, Smart Package Generation and Explainable Recommendation work in conjunction with each other. Therefore, rather than developing the system completely at once, it is more appropriate to develop it in parts and test them and make the necessary changes.

This approach also suits the nature of the product. The proposed system is an intelligent system that provides vendor compatibility, risks, venue suitability and package recommendations using data such as event requirements, vendor information and venue information. While developing these, it may be necessary to improve the recommendation logic, criteria and system features. Therefore, it is important to develop the system in stages, testing the results at each stage and making improvements.

The main users of this system can be identified as event planners, event organizers and clients. These user needs and the recommendations, explanations, and usability they expect from the system can change or improve during the development process based on feedback. The Agile approach allows for such user feedback to be received and the necessary changes to be made in the next development iteration.

The development can be carried out in stages in the project as follows. First, vendor information analysis and vendor selection functions can be developed, then vendor compatibility and risk analysis can be added. Then, venue suitability validation, multi-vendor package generation, and recommendation explanation features can be developed. At the end of each stage, the system can be tested, and the system can be improved based on the errors, conflicts, and user feedback that are identified.

Accordingly, the Agile Development Approach with Iterative Development has been selected for this research project, considering the complexity of the product in this project, the changing needs of the users, the interdependent components of the system, and the need for constant testing and improvement.

4 HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Proposed Solution and its Components

The aim of this project is to integrate the information required for planning an event into a single system and provide solutions that are suitable for the event. For this, the system consists of several components, and each component fulfills a different need in the event management process.

My individual components are Vendor Intelligence and Smart Package Recommendation. In this component, factors such as the requirements of the event, budget, vendor details, vendor reliability, availability, compatibility between vendors, operational risks, and venue suitability are considered. Accordingly, vendors that are suitable for the event are identified, and a suitable vendor package is recommended by connecting several vendors that are compatible with each other. When providing that recommendation, it is also explained to the user why it was selected.

The main parts of the component that I developed are Vendor Intelligence, Vendor Compatibility and Risk Analysis, Venue Suitability Validation, Multi-Criteria Recommendation, Smart Package Generation and finally Explainable Recommendation.

Vendor Intelligence extracts and structures relevant data of vendors

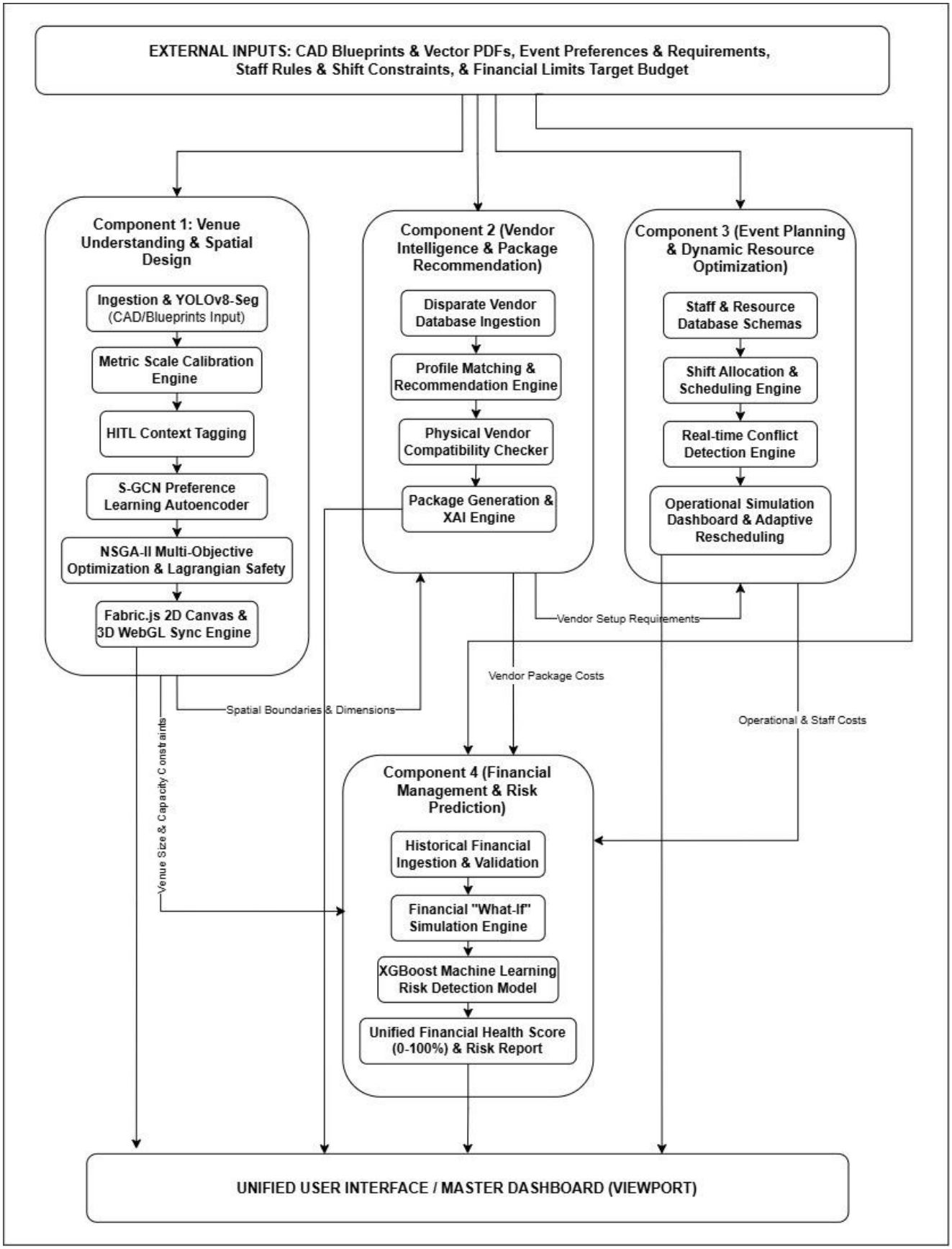


Figure 4.1 System Diagram of Components

4.2 System diagram of components

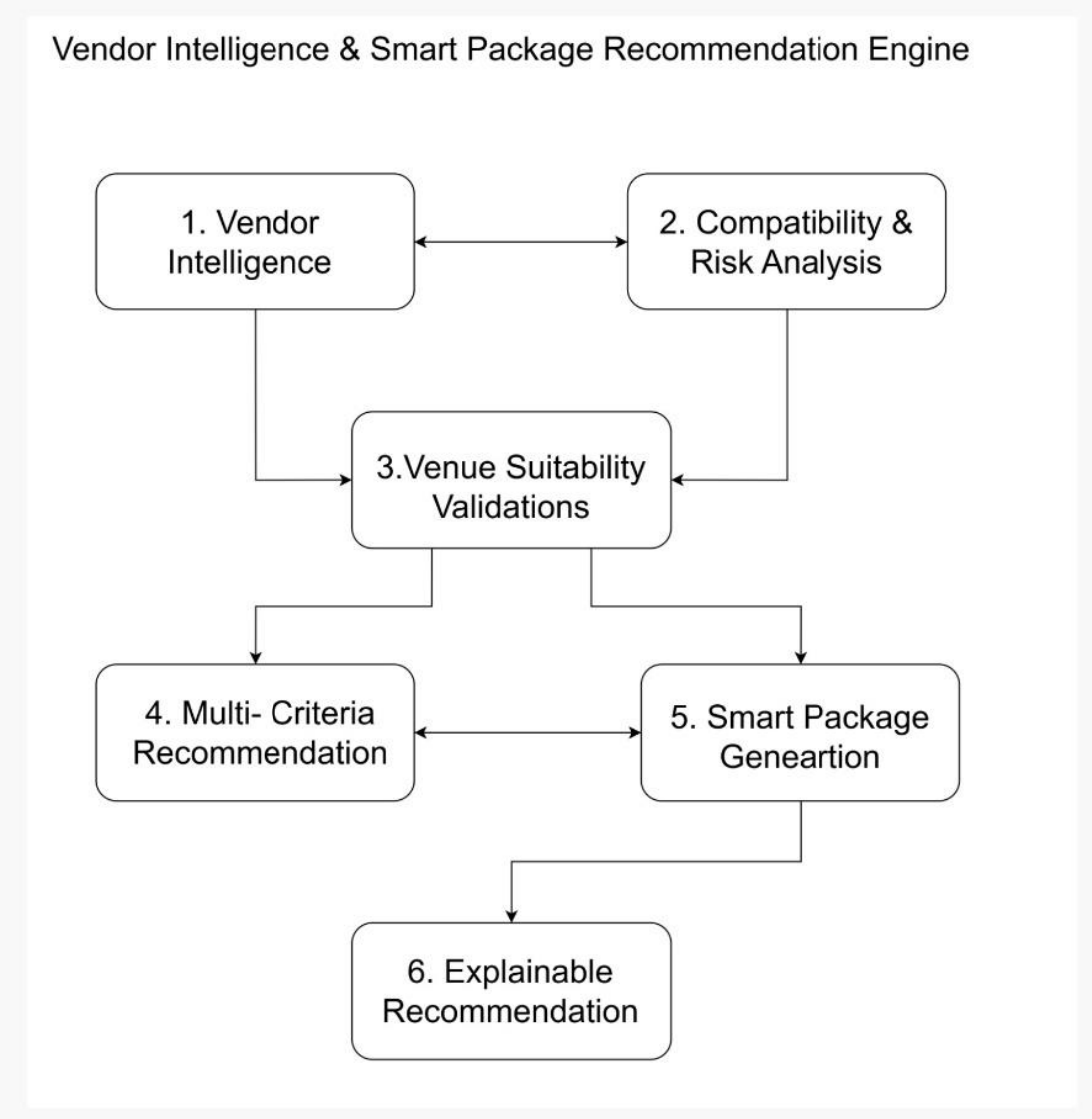


Figure 4.3 System diagram of individual component

4.3 Highlight the individual component and its parts

My individual component in this project is Vendor Intelligence and Smart Package Recommendation. The main objective of this component is to identify vendors that match the needs and budget of the event, consider whether those vendors are compatible with each other and are suitable for the venue where the event is held, and finally recommend a suitable vendor package.

This component consists of several main parts.

Vendor Intelligence

This part collects and analyzes information about vendors. Information such as vendor category, services provided, price, rating, availability and reliability are used to identify vendors that are suitable for the event.

Vendor Compatibility and Risk Analysis

This part checks whether the vendors being considered for selection are compatible with each other. This also includes identifying resource conflicts, service conflicts and operational risks that may arise when multiple vendors work together on the same event.

Venue Suitability Validation

This part checks whether the venue selected for the event is suitable for the selected vendors. Venue capacity, available facilities, space requirements, and vendor operations are considered here.

Multi-criteria Recommendations

In this section, the vendors are evaluated against criteria including needs of the event, budget, dependability of the vendor, availability, compatibility, and suitability of the venue. Therefore, the most suitable vendors for the event are identified.

Smart Package Generation

Instead of providing only individually suitable vendors, this section creates a complete vendor package by combining several vendors that are compatible with each other. Factors such as budget, event requirements, vendor compatibility, risks, and venue suitability are considered together when creating the package.

Explainable Recommendation

Finally, the system clearly shows the user why the vendor package recommended by the system was chosen. For example, the key factors that influenced the recommendation can be shown, such as the package matching the event requirements and budget, good compatibility between vendors, and being suitable for the venue.

Accordingly, by working together, the Vendor Intelligence and Smart Package Recommendation component can identify vendors that are suitable for an event, check the compatibility and risks between them, consider venue suitability, and ultimately provide the user with a suitable vendor package with clear reasons.

4.4 Overall Integration

The system is designed in such a way that the components of the overall system work together. After the event requirements, budget, and other requirements provided by the user are entered into the system, they are provided to the required components.

My component requires event requirements, vendor information, and venue-related information as its main input. Vendor selection is done using event information such as event type, guest count, required services, and budget. Vendor details consider factors such as services, price, availability and reliability. Information such as the venue's capacity, facilities and operational limitations are used to check whether the venue is suitable for the selected vendors.

Then, compatibility between vendors, possible conflicts and operational risks are checked, and a suitable vendor package is recommended considering all these factors. That package and the reasons for the recommendation are provided to the user through the overall system.

It is planned to use API-based integration and a shared data structure for data exchange between components. This allows the required data to be obtained from one component and used for processing in another component. Ultimately, all components will be connected to provide a complete solution for event planning.

4.5 Responsibility Boundary of My Component

My individual responsibility is for the tasks related to the Vendor Intelligence and Smart Package Recommendation component. Accordingly, analyzing vendor information, identifying vendor compatibility and risks, checking venue suitability, connecting multiple vendors together and recommending a suitable package, and explaining that recommendation are my main responsibilities.

Developing the main functions of other group members' components is not my responsibility. However, obtaining the necessary inputs from those components to my component and providing the necessary outputs from my component are done as part of the overall system integration.

4.6 Individual Responsibility Boundary

Activity / Deliverable	My Responsibility	Shared Responsibility	Other Member
Analyzing events and vendor information	✓		
Identifying vendor compatibility and risks	✓		
Checking venue suitability	✓	✓	
Multi-criteria vendor recommendation	✓		
Creating a smart vendor package	✓		
Explanation of the reasons for the recommendation	✓		
Data/API integration between components		✓	✓
Overall system integration and testing		✓	✓
Key functions of other components			✓

Table 4.5: Individual Responsibility Boundary Table

This architecture shows that my component does not act as a separate vendor selection system, but rather as a component that connects with other components of the Integrated Event Management System to recommend a complete vendor package that is compatible with the event.

5 USER REQUIREMENTS

5.1 Identifying User Requirements

In identifying user requirements for this component, the event planning and event management process was considered. When planning an event, activities such as selecting several vendors, checking their price, availability, reliability and services are carried out. It was identified that when selecting several vendors together, attention should be paid to their compatibility, suitability for the venue and possible operational problems.

Stakeholder discussions, observation of existing event planning practices and domain knowledge were used to identify these user requirements. By considering how vendors are selected and compared in the event planning process, the main needs of users were identified. The opinions of people with knowledge of event management and existing vendor selection practices were also considered.

5.1.1 Key Requirements of Users

The main users of this system can be identified as event planners, event organizers and clients. The key requirements they need are as follows.

- The system should be able to enter event requirements such as event type, guest count, required services and budget.
- It should be possible to obtain information about vendors in various vendor categories such as catering, photography, decoration, entertainment and audio-visual.
- It should be possible to identify suitable vendors by considering factors such as price, rating, availability, reliability and services provided by a vendor.
- When selecting several vendors together, it should be possible to check their compatibility.
- It must be feasible to spot potential conflict of resources and operation risks that might come up due to multiple vendors being used at once.
- It must be feasible to find out if the chosen vendors are suitable to use at the event venue considering the capacity, availability of space, and facilities needed.
- It should be possible to combine several vendors together and obtain a suitable package considering the budget and event requirements.
- Users should be able to clearly understand the reason for the package recommendation provided by the system.

- If necessary, other suitable vendor packages or alternative recommendations should also be available.
- The system's recommendations and results should be easy to understand and simple to use.

5.1.2 Transforming User Requirements to System Requirements

Supply information - about the events the system must be able to input the event type, number of guests, services needed, and budget.

Obtaining vendor information - such as vendor category, services, price, rating, availability and reliability - should be available to the system.

Identifying suitable vendors - It should be possible to identify and rank suitable vendors according to the event requirements and selected criteria.

Checking vendor compatibility - It should be possible to check service, resource and operational compatibility between selected vendors.

Identifying risks and conflicts - It should be possible to identify operational risks and conflicts that may arise when using multiple vendors together.

Checking venue suitability - It should be possible to check venue suitability by considering venue capacity, facilities, available space and vendor requirements.

Obtaining a suitable vendor package - It should be possible to combine compatible vendors and generate a package considering budget, requirements, availability, reliability, compatibility, risks and venue suitability.

Knowing the reasons for the recommendation - It should be possible to explain why the selected package was recommended with factors and reasons.

Obtaining alternative options - It should be possible to provide other suitable vendor package alternatives when necessary.

Using an easy system - There should be a simple interface where users can easily enter event details and get recommendations and explanations.

5.1.3 Summary

Accordingly, the user requirements require users to obtain a suitable vendor package by considering several compatible vendors according to the event requirements and budget, rather than searching for individual vendors separately. The vendor compatibility,

operational risks, venue suitability and reasons for the recommendation of that package also need to be clearly shown. Based on these findings, the system requirements of the proposed component have been identified.

5.2 Translation of Findings into System Requirements

5.2.1 Functional requirements

ID	Name	Functional Requirement
FR1	**Retrieve Event Details**	The system should be able to retrieve event type, guest count, required services, budget and other requirements related to the event.
FR2	**Retrieve Vendor Details**	The system should be able to retrieve and use vendor information such as vendor category, services, price, rating, availability and reliability.
FR3	**Identify Suitable Vendors**	The system should be able to identify and evaluate vendors that are suitable for the event based on the given event requirements and budget.
FR4	**Check Vendor Compatibility**	The system should be able to check whether the vendors being considered for selection are compatible with each other.
FR5	**Identify Risks and Conflicts**	The system should be able to identify resource conflicts, service conflicts and operational risks that may arise when multiple vendors work together.
FR6	**Venue Suitability Evaluation**	The system should provide the functionality to determine whether the venue is suitable for the chosen vendors, considering such aspects as venue capacity, facilities, space, and vendor requirements.
FR7	**Vendor Evaluation and Ranking**	The system should provide the ability to evaluate and rank vendors in terms of price, reliability, availability, compatibility, venue suitability, and other event-related requirements.
FR8	**Creating a Smart Vendor Package**	The system should be able to create a suitable vendor package by combining multiple vendors that match

		the event requirements and budget and are compatible with each other.
FR9	**Providing Vendor Package Recommendation**	The system should be able to provide the most suitable vendor package for the event to the user as a recommendation.
FR10	**Providing reasons for the recommendation**	The system should clearly show the user why the recommended vendor package was chosen and why it is suitable for the event.
FR11	**Providing Alternative Vendor Options**	The system should be able to show other suitable vendor options or packages in addition to the recommended package, when necessary.
FR12	**Displaying Recommendation Results**	The system should display results such as selected vendors, package details, estimated cost, compatibility status, venue suitability and identified risks in a way that is easy for the user to understand.
FR13	**Re-sending Recommendation**	When the user changes the event requirements, budget or vendor preferences, the system should be able to re-send vendor recommendations and the package based on those changes.

Table 5.5 Functional requirements Table

5.2.2 Non-functional requirements

• **Performance** – The system should be able to provide vendor package recommendations within 5 seconds of event details, even if they contain a large amount of vendor and venue data.

• **Recommendation Accuracy** – The vendor package recommendations provided by the system should have a recommendation accuracy of at least 80% when compared to the predetermined correct results.

• **Response Time** – After the user provides the event requirements, budget, and other necessary information, the system's response time to provide suitable vendors and the final vendor package recommendation should not exceed 5 seconds.

• **Security** – The event details provided by the user and the vendor information in the system should be protected from unauthorized users. Unnecessary storage of user data should also be avoided.

• **Reliability** – In cases where vendor data is incomplete, a vendor is unavailable, or a suitable vendor combination is not available, the system should not crash, and the user should be provided with an appropriate message or alternative option.

• **Usability** – The system interface should be simple and clear so that even a user without technical knowledge can enter event details, obtain vendor recommendations, and understand the reasons given for the recommendation.

• **Scalability** – The system's core functions should continue to function as the number of vendor and venue records increases, and the response time should not decrease significantly as the amount of data increases.

• **Accessibility** – The system's text, buttons, input fields, and recommendation results should be clearly displayed, and the system should be easy to use for a variety of users.

6 COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The main target market of this system is the event planning and event management sector. Wedding planners, event management companies, corporate event organizers, hotels and other event organizers can be identified as the main customers here. The system can also be used by clients who want to select vendors for their event.

6.2 Value Proposition

The main value of this system is that instead of searching for vendors one by one, several vendors that match the requirements and budget of the event are considered together and provided with a suitable vendor package. Also, the vendor selection process can be facilitated by checking the compatibility, operational risks and venue suitability between vendors and showing the reasons for the recommendation to the user.

6.3 Revenue Model

A subscription-based model can be used to generate revenue from this system. Monthly or yearly subscription plans can be provided for event planners and event management companies. In addition, a system of establishing partnership agreements with vendors and receiving a commission on referrals or successful bookings can also be used.

6.4 Cost Estimation

Developing the system involves costs for software development, database hosting, cloud services, maintenance, security, and system updates. While initial development and testing may require more expenditure, there will be recurring costs for hosting and maintenance once the system is put into commercial use.

6.5 Pricing Strategy

Different pricing plans can be introduced depending on the customer type and required features. For example, a basic plan can be provided for individual users, a standard plan for event planners and small businesses, and a premium plan for large event management companies and hotels. This can provide pricing options that are suitable for different customer groups.

6.6 Competitive Advantage

The main competitive advantage of this system is that it considers several vendors together and provides a package-level recommendation instead of recommending vendors individually. The system differentiates itself from other typical vendor search systems by providing recommendations that consider vendor compatibility, operational risks, venue suitability, and budget, and by clearly stating the reasons for the recommendations.

6.7 Intellectual Property Considerations

The developer-created parts of this system, such as source code, recommendation logic, system architecture, and software design, may be protected as intellectual property. Trademarking the system's name and logo may also be considered as needed. Furthermore, when using third-party vendor data, images, and other external content, applicable copyright, licensing, and data usage requirements must be adhered to.

7 BUDGET AND JUSTIFICATION

7.1 Hardware Costs

The developer's existing computer/laptop can be used to develop and test the system. If additional hardware is required, a computer with sufficient RAM and processing power for development and testing, and external storage as needed, may be required.

Justification

Sufficient hardware resources are required to perform tasks such as vendor data processing, system development, testing, and recommendation generation.

7.2 Software Licenses

Since programming languages, development frameworks, database tools, and other open-source software tools can be used to develop the system, the cost of software licensing can be minimized. If paid software or other third-party services are required, license fees for them should be considered.

Justification

Development tools and supporting software are required to develop and test the system, and using open-source tools can reduce unnecessary software license costs.

7.3 Cloud Services

When deploying the system for commercial use, there may be costs for cloud services such as cloud hosting, database storage, and backup services. These costs can also increase as the number of users and vendor data on the system increases.

Justification

Cloud services are required to access the system online, store vendor and venue data, backup data, and maintain the system reliably.

7.4 Human Effort Cost

The time and effort of developers, system designers, and testers is required to design, develop, test, and maintain the system. This includes the time spent on requirements analysis, vendor data preparation, recommendation logic development, testing, and system improvements.

Justification

Skilled human effort is required to accurately develop key features of the system such as recommendation functions, vendor compatibility checking, venue suitability validation, and package generation.

7.5 Overall Expense Justification

All these costs are the initial costs required for the development, testing, and future deployment of the system. The costs can be reduced by using existing hardware and open-source software when developing the initial prototype. When moving the system to a commercial level, separate costs for cloud hosting, maintenance, security, and additional development must be considered.

8 WORK BREAKDOWN STRUCTURE (WBS)

8.1 Task Breakdown Per Member

Member 1: Fernando W S V (Venue Understanding & Spatial Event Design Intelligence)

- I**ngestion & Measurement:** Develop computer vision system (YOLOv8-Seg) to process blueprint images into floorplans, and text-recognition systems (CRNN/Tesseract OCR) to automatically measure real-world length/width of rooms.

- **Interactive Design & Style Learning:** Design interactive screen canvas to allow planners to snap permanent structures (walls, pillars), and tag moveable objects, with graph neural networks (S-GCN) to learn one's design style.

- **Layout Optimization & Explanations:** Build optimization engine (NSGA-II) to find ideal event layout subject to safety regulations, with built-in AI reasoner to explain item placements.

- **3D Visual Twin & Safety Guardrails**: Develop 2D to 3D projection engine for digital twin visualization; implement real-time pathfinding safety analytics (A*) to enforce minimum clearance (1.8m) of emergency exits.

Member 2: Ahinsi K R N (Vendor Intelligence & Package Recommendation Engine)

- **Data & Ingestion Layer:** Design database schemas, validation rules, and ingestion scripts for disparate vendor databases.

- **Core Recommendation Engine**: Build recommendation engine that learns from vendor pricing/service profiles and matches user needs.

- **Compatibility & Risk Analysis:** Design compatibility engine that cross-references with Component 1 (Venue Digital Twin) to ensure physical compatibility of chosen vendors with venue.

- **Package Generation & XAI:** Design system for generating bundled service packages with natural language and visual explanations.

Member 3: Gamage J H A D N (Event Planning & Dynamic Resource Optimization

- **Scheduling & Allocation:** Design database schemas and scheduling engine for allocating staff/resources to shifts.
- **Conflict Detection:** Build real-time conflict detection engine to identify scheduling/resource allocation issues.
- **Operational Digital Twin & Simulation:** Build real-time operations dashboard and event-planning simulation engine.
- **Adaptive Re-scheduling:** Design adaptive rescheduling algorithms with dynamic financial recalibration.

Member 4: Weerasekara T P (Financial Management & Risk Prediction / Unified Financial Health Score)

- **Financial Ingestion & Validation**: Design historical financial records ingestion and validation systems, with field-specific constraints relating expenses to venue size.
- **"What-If" Simulation Engine:** Build financial "what-if" engine to explore real-time financial scenarios.
- **Risk Detection Model:** Build machine learning model (XGBoost) to identify outliers and potential financial risks.
- **Scoring & Reporting**: Design financial health score (0-100%) algorithm and risk assessment report visualization.

8.2 Timeline Alignment

Project Phase	Timeline Milestone	Core Focus & Member Task Breakdown	Inter-Component Dependencies & Data Flow
Research & Ingestion	Month 1 – 2	**All Members:** • **Member 1:** Blueprint CAD ingest and OCR parsing. • **Member 2:** Vendor database schema & profiling.	Independent data layer setups and baseline schema design across all four modules.

		• **Member 3:** Activity & resource data structuring. • **Member 4:** Historical budget & financial limit ingestion.	
Core AI & Engine Development	Month 3 – 4	**All Members (Active Parallel Development):** • **Member 1:** YOLOv8-Seg, S-GCN style learning, & NSGA-II optimization setup. • **Member 2:** Vendor multi-criteria matching & package generation logic. • **Member 3:** Intelligent scheduling engine & disruption simulation algorithms. • **Member 4:** XGBoost financial risk detection & health scoring algorithms.	Core AI models trained independently and standardized API endpoints for spatial-data consumption.
Cross-Component Integration & Digital Twin Linking	Month 5 – 6	**All Members (Unified Integration):** • **Member 1:** Finalizing WebGL Digital Twin viewport and A^* pathfinding guardrails. • **Member 2:** Integrating vendor packages with Member 1's spatial layout and footprint bounds. • **Member 3:** Syncing operational resource schedules and disruption alerts with the spatial environment. • **Member 4:** Linking financial budget constraints, cost signals, and rate cross-checks with	Critical Integration Point: Members 2, 3, and 4 actively consume spatial data, boundary coordinates, and safety constraints from your component (Component 1) to validate vendor placements, resource paths, and venue limits.

		Component 1's spatial digital twin.	
Evaluation & Testing	Month 7 – 8	**All Members:** • End-to-end system testing and cross-component API stress testing. •Domain expert validation sessions, XAI report generation, and final report/presentation binding.	Complete end-to-end verification of the unified event platform.

Table 8.2 Timeline Alignment Table

9 GANTT CHART

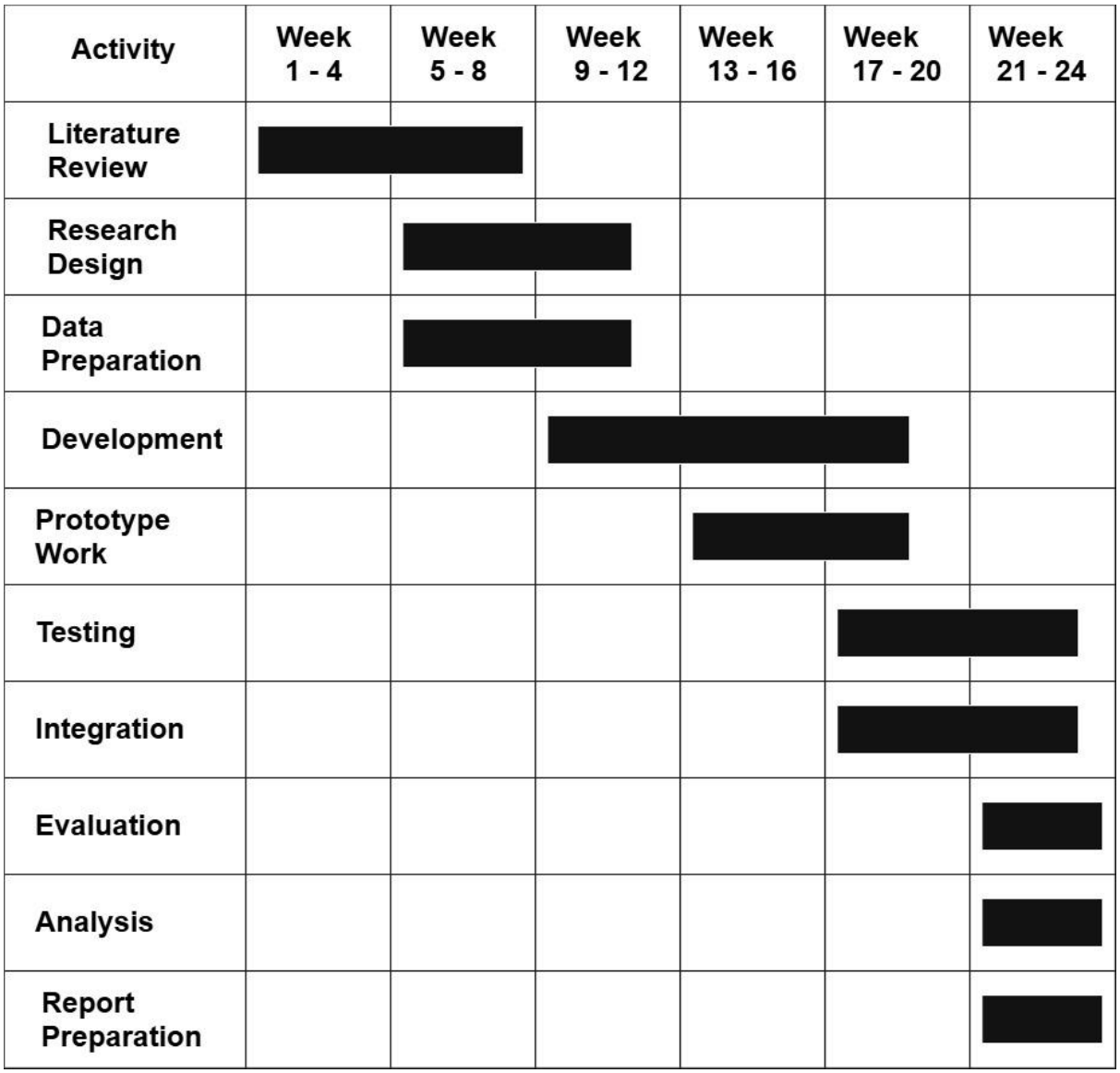


Figure 9 : Gantt Chart

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11 APPENDICES

Appendix 1 – Problem Analysis

While organizing an event, various vendor services like catering, photography, decoration, entertainment, and audio-visual services are required. At present, when choosing these vendors, the event planner or the customer looks at each vendor separately and compares the price, rating, availability, etc. to make decisions.

But when choosing several vendors for the same event, it is difficult to consider things like compatibility between them, service conflicts, operational risks and compatibility with the venue at the same time. For example, there may be a conflict between the services of two selected vendors, and the equipment or setup space required by one vendor may not be sufficient in the venue.

Due to this situation, problems such as spending more time on the vendor selection process, selecting unsuitable vendors and operational problems during the event may occur.

Accordingly, in this research, a Vendor Intelligence & Smart Package Recommendation System has been proposed to identify compatible vendors by considering vendor information, event requirements and venue information and to recommend a vendor package suitable for the event.

Appendix 2 — Mandatory Proposal Evidence and Decision Log

Date	Claim / Problem	Evidence Collected	Alternatives Considered	Decision Made	Supervisor Discussion
2026.07.06	When organizing an event, the vendor selection process becomes complicated due to the need to select several vendors such as catering, photography, decoration, entertainment, etc. separately.	Information on existing vendor platforms and vendor selection approaches was studied.	Manual vendor selection, using existing vendor marketplace, developing an integrated vendor recommendation system.	It was decided to develop a system that analyzes several vendors simultaneously and recommends a suitable vendor package according to the event requirements.	The research problem was clearly defined, and the scope of the research component was discussed with the Supervisor.

2026.07.10	When selecting a vendor, it is not possible to identify the most suitable vendor by using only price or rating.	Literature and MCDM research papers on supplier/vendor selection were studied.	Using price only, using price and rating, using several criteria simultaneously.	It was decided to use several criteria such as price, rating, reliability, availability, compatibility and venue suitability.	Recommendation criteria and their importance were discussed with the Supervisor.
2026.07.25	Selecting vendors separately makes it difficult to identify service conflicts and operational problems between vendors.	Vendor selection literature and event planning requirements were studied.	Selection approach included individual vendor selection, simple rule-based filtering, compatibility and risk analysis.	It was decided to include a Vendor Compatibility & Risk Analysis module in the system.	Feedback was obtained from the Supervisor on vendor compatibility and operational risk criteria.
2026.07.29	Before selecting a vendor, it is necessary to check whether the vendor has the required facilities and space at the selected venue.	Information on venue capacity, facilities, available space and vendor operational requirements was studied.	Vendor selection without regard to venue suitability, basic venue filtering, venue suitability validation.	It was decided to include Venue Suitability Validation as part of the recommendation process.	Venue suitability criteria and venue-related validation process were discussed with the Supervisor.
2026.07.30	When the system provides a recommendation, the user should be able to understand why that recommendation was selected.	Literature on Explainable AI and recommender systems was studied.	Providing only the recommendation result, providing a simple reason, Explainable AI-based explanation.	It was decided to include an Explainable Recommendation / XAI component in the system.	The need to make the recommendation explainable and the content of the explanation were discussed with the Supervisor.
2026.08.01	When selecting multiple vendors for an event, multiple criteria need to be considered simultaneously.	Research studies on MCDM and multi-criteria vendor selection were studied.	Rule-based selection, simple ranking, Multi-Criteria Decision-Making approach.	It was decided to use the Multi-Criteria Decision Making (MCDM) approach for the recommendation process.	The MCDM approach was discussed with the Supervisor on how it fits into the research component.

Table 11.2 : Mandatory Proposal Evidence and Decision Log Table

Appendix 3 — AI Use Disclosure

AI Tool	Purpose of Use	What Was Generated/Assisted	How Output Was Verified	Final Student Contribution
ChatGPT	To make the research proposal writing clearer	I received help to improve grammar, sentence structure and paragraph organization	I compared the suggestions given with my research scope, objectives and ideas and made the necessary changes	I selected, edited and finalized the final content
ChatGPT	To design the structure and presentation of the proposed system	Identified ideas and structures for system diagrams, data flows, research methodology flowchart, and WBS	I tested these ideas against my actual system architecture and project requirements	I determined the final system structure, component relationships, and diagram content
ChatGPT	To better understand the research concepts and proposal,	Explanations and suggestions were obtained on issues such as research methodology, research gaps, system components, workflows and WBS	Those suggestions were compared and checked with my project requirements, research objectives and system scope	I was responsible for selecting the necessary issues and making the final research decisions

Table 11.3 : AI Use Disclosure Table

Appendix 4 – Work Breakdown Structure (WBS)

1. Vendor Intelligence & Smart Package Recommendation System

1.1 Project Planning

- 1.1.1 Research Problem Identification
- 1.1.2 Research Objectives Definition
- 1.1.3 Project Scope Definition
- 1.1.4 Project Planning

1.2 Literature Review & Requirement Analysis

- 1.2.1 Literature Review
- 1.2.2 Research Gap Identification

- 1.2.3 User Requirement Identification
- 1.2.4 System Requirement Definition

1.3 Data Collection & Preparation

- 1.3.1 Vendor Data Collection
- 1.3.2 Venue Data Collection
- 1.3.3 Event Data Definition
- 1.3.4 Data Cleaning
- 1.3.5 Data Validation and Formatting

1.4 System Design

- 1.4.1 System Architecture Design
- 1.4.2 Vendor Intelligence Design
- 1.4.3 Compatibility and Risk Analysis Design
- 1.4.4 Venue Suitability Design
- 1.4.5 Recommendation Design
- 1.4.6 Package Generation Design
- 1.4.7 Explainable Recommendation Design

1.5 System Development

- 1.5.1 Vendor Intelligence Development
- 1.5.2 Compatibility and Risk Analysis Development
- 1.5.3 Venue Suitability Validation Development
- 1.5.4 Multi-Criteria Recommendation Development
- 1.5.5 Smart Package Generation Development
- 1.5.6 Explainable Recommendation Development

1.6 Integration & Testing

- 1.6.1 Component Integration
- 1.6.2 Functional Testing
- 1.6.3 Compatibility Testing
- 1.6.4 Venue Suitability Testing
- 1.6.5 Recommendation Testing
- 1.6.6 Performance Testing

1.7 Evaluation

- 1.7.1 Recommendation Accuracy Evaluation
- 1.7.2 Vendor Compatibility Evaluation
- 1.7.3 Venue Suitability Evaluation
- 1.7.4 Conflict Identification Evaluation
- 1.7.5 Explanation Quality Evaluation
- 1.7.6 Response Time Evaluation
- 1.7.7 Usability Evaluation

1.8 Documentation & Final Submission

- 1.8.1 Results Analysis
- 1.8.2 Final Report Preparation
- 1.8.3 Presentation Preparation
- 1.8.4 Final Submission

Appendix 5 – Literature Comparison

In reviewing previous research relevant to this research, research in various fields such as vendor selection, multi-criteria decision making, machine learning, risk analysis, explainable recommendation and venue recommendation was identified.

Research Area	Focus of Previous Research	Identified Limitations
Vendor Selection	Selecting vendors based on price, quality, rating and other criteria	Not fully considering compatibility between multiple vendors
Multi-Criteria Decision Making	Selecting the best alternative using multiple criteria	Limited integrated approach directly applied to event vendor packages
Machine Learning	Using predictive models for vendor selection and decision support	Limited linking of venue suitability and vendor compatibility
Risk-Aware Vendor Selection	Considering vendor reliability and operational risk	Not fully integrating with event-specific multi-vendor package recommendation
Explainable Recommendation	Reasoning for a Recommendation	Limited explanation of vendor compatibility, venue suitability, and package-level decisions
Venue Recommendation	Identifying a suitable venue for an event	Venue suitability not linked to vendor selection and package generation

Table 11.5 : Literature Comparison Table

The above literature review pointed out the shortcomings in current methods and the necessity for a unified approach that considers all aspects like vendor selection, compatibility, risk, suitability of venues, multi-criteria decision-making, and explainability.

Appendix 6 – Gantt Chart

Visualization of the Gantt Timeline: The following Gantt diagram shows the process of research and development for the "Vendor Intelligence & Smart Package Recommendation System," which is scheduled to last 24 weeks. It highlights the different activities that include literature study, research planning, data processing, development of the system, development of the prototype, testing, integration, evaluation, results analysis, and preparation of the report.

Nethmi Ahinsi

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Assignment 11

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